Lab 4

Windows Forensics

In this lab we are going to do some analysis on Windows OS.

Firstly, we will analyze MAC timestamp, then we will change the Timestamp as a way to deceive the investigators.

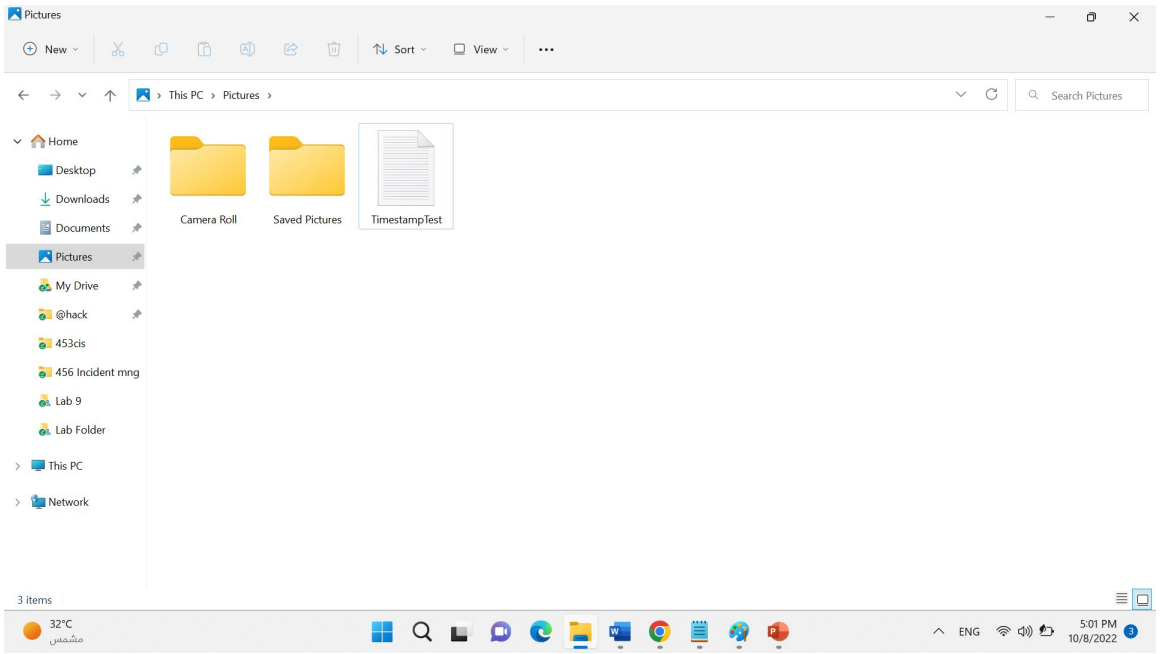
Secondly, analyze Prefetch file in Windows.

## Task 1: Timestamp Analysis:

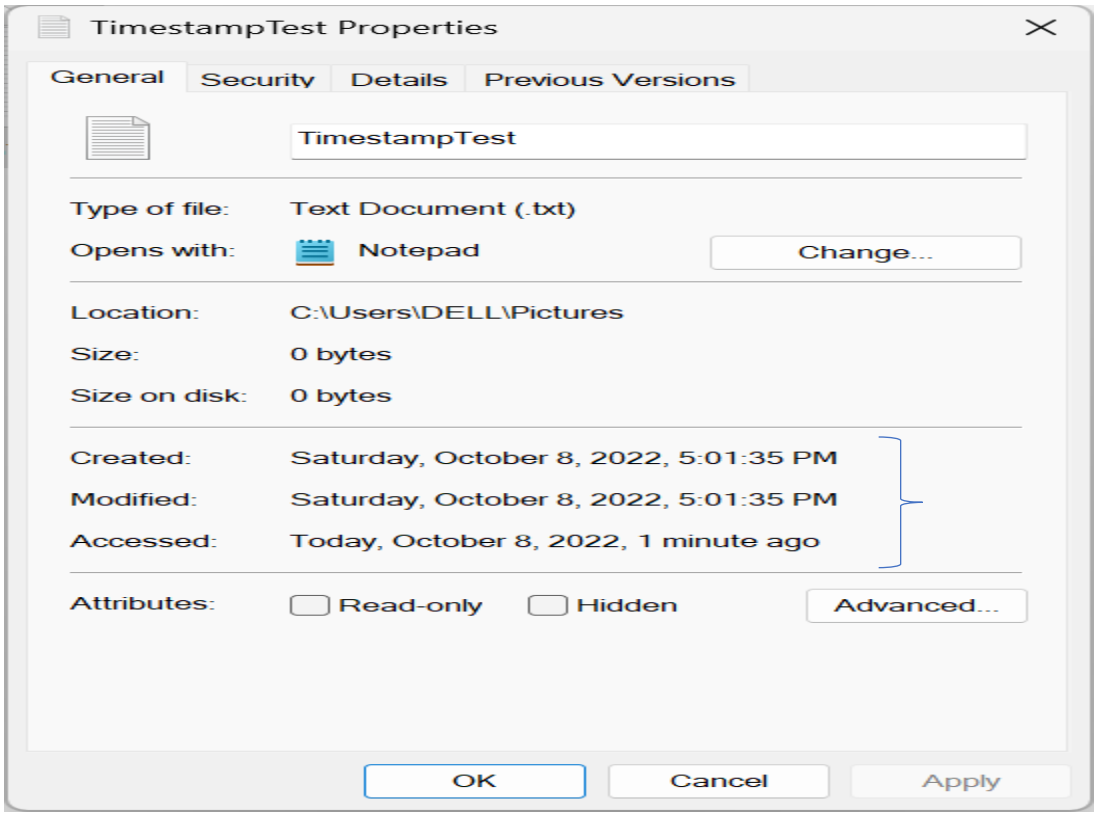
In this task we are going to examine timestamp for a new file you will create.

| Operation | Modified | Accessed | Birth (Created) |
|---|---|---|---|
| File Create | Yes | Yes | Yes |
| File Modify | Yes | No | No |
| File Copy | No (Inherited) | Yes | Yes |
| File Access | No | No* | No |

1- Create a new file for example in Pictures folder.

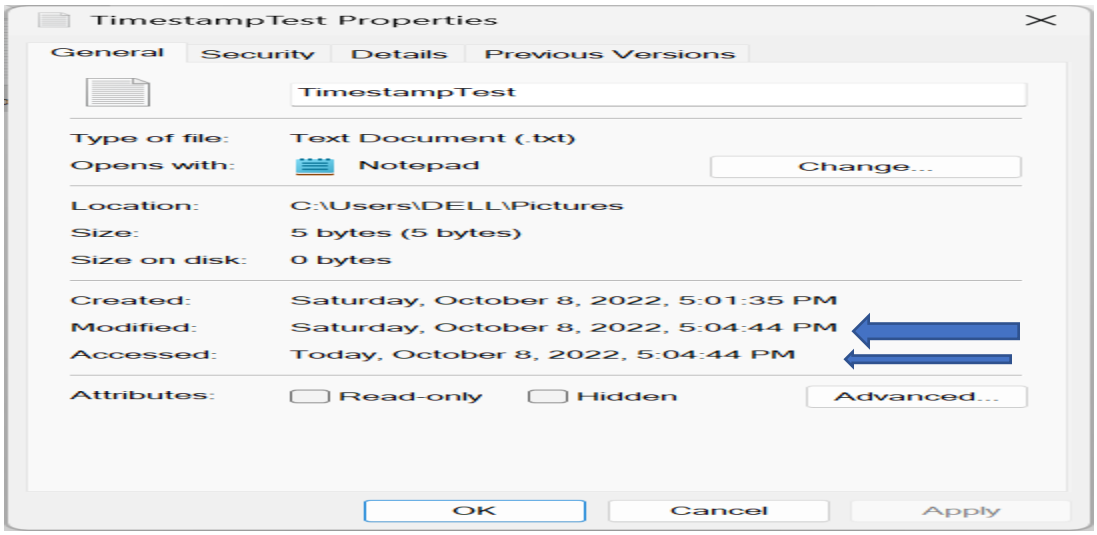



2- Check File Timestamp, by right click>>Properties:



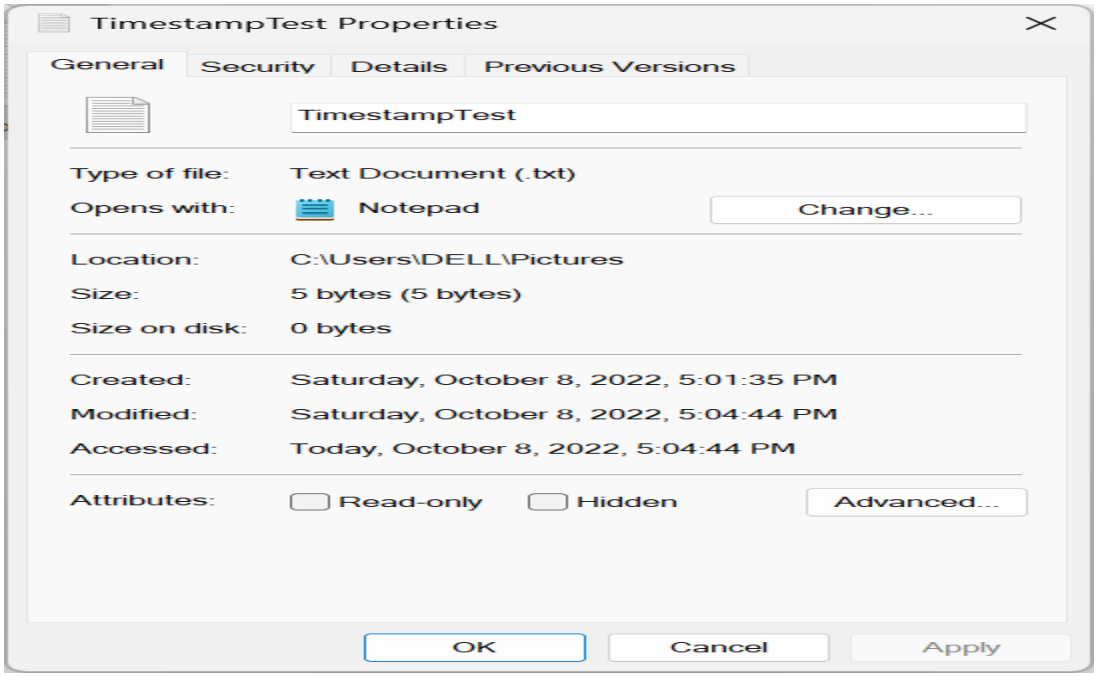

You will find all MAC (Modified, Accessed, Created) were changed accordingly.

3- Do some changes (Modify) in the file, and re-check Timestamp again.



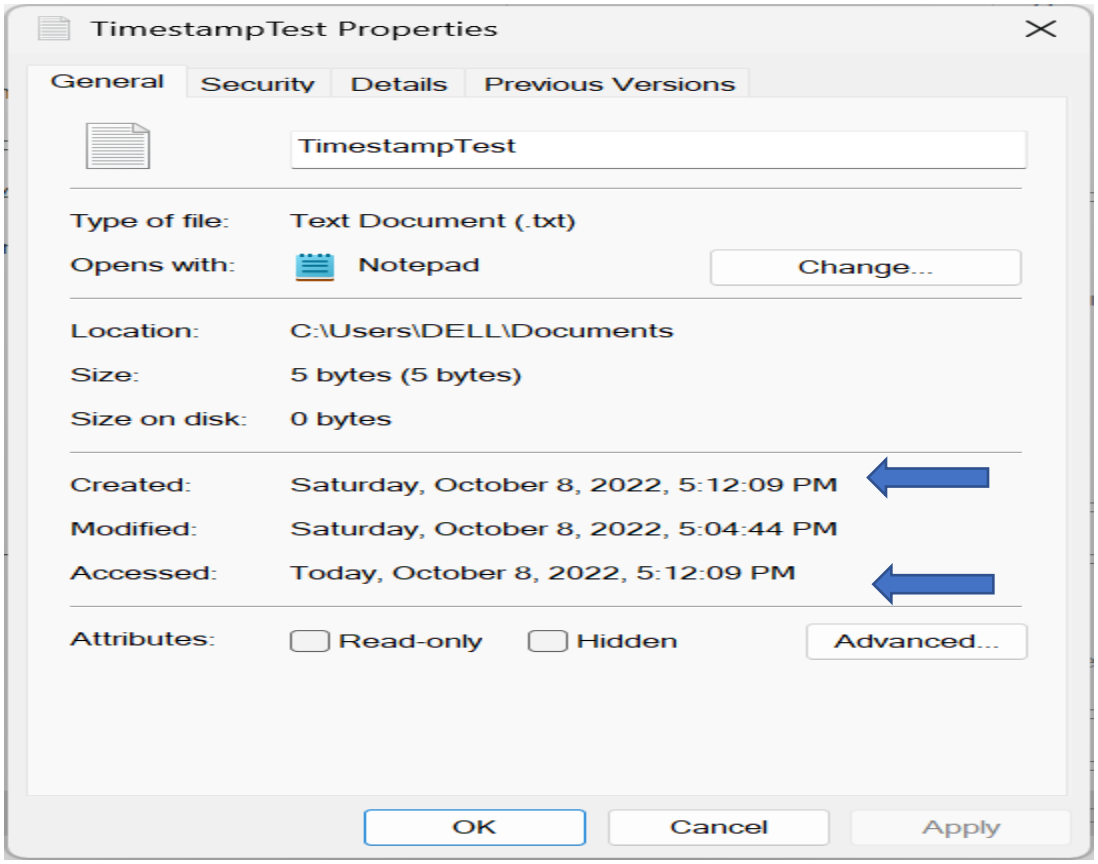

Please not here that Created is not changed, and Accessed is changed but in some systems it might now be changed immediately, some times it takes 1 hour to be changed.

4- Try now to only access the file, and observe timestamp:




In my case it is not changed (although I accessed at 05:09pm).

5- Finally, make a copy of the file and observe timestamp of the new copied file:



Task 2: Use Timestomp to change timestamp (to hide real timestamp from incident response team).

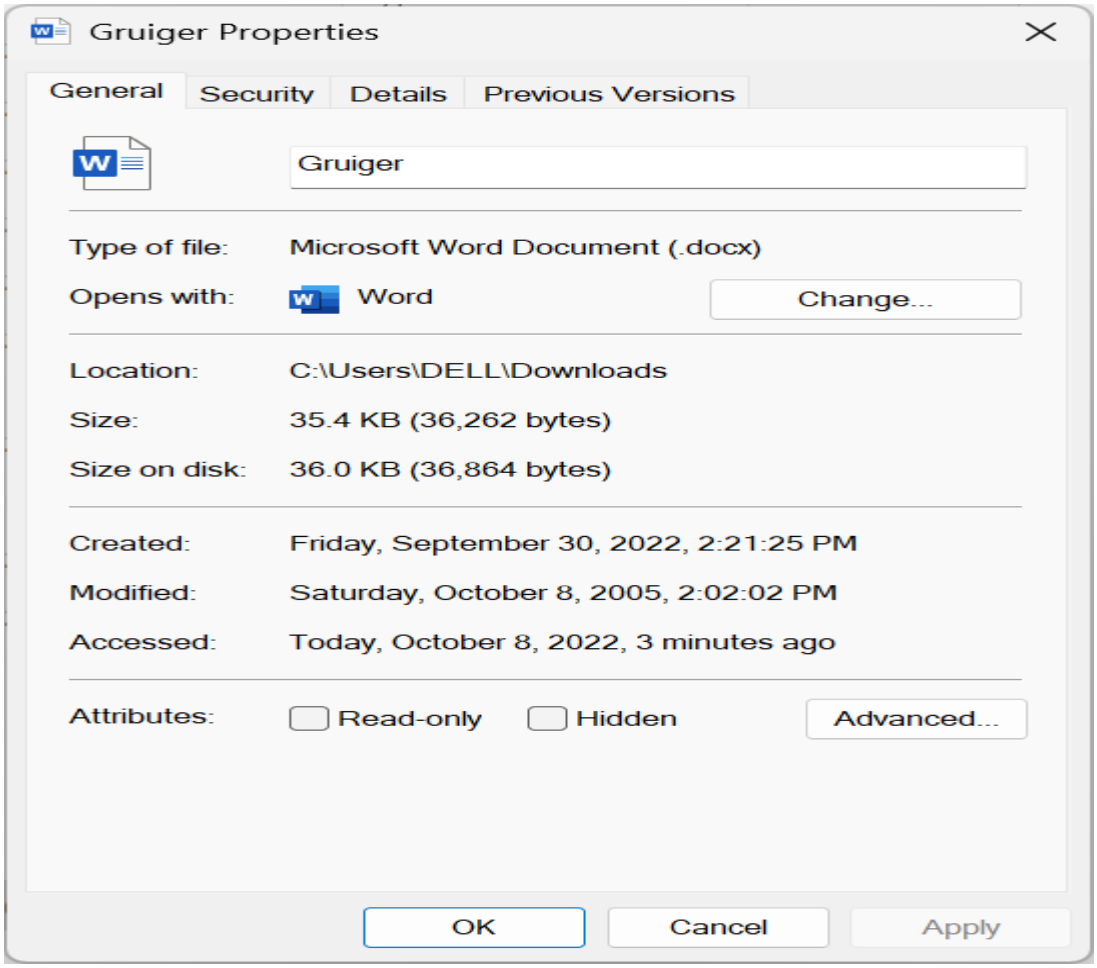
1- Create a new file in Downloads folder, and observe MAC timestamp.
2- Download tool timestomp.exe.
3- Open the terminal and change directory (cd) to the same folder where Tiestomp.exe is installed. (in my case in Downloads Folder).

```
C:\Users\DELL>cd /Users/DELL/Downloads
```

4- Type in Terminal the command below to change the timestamp of modified time:

```
C:\Users\DELL\Downloads>timestomp.exe Gruiger.docx -m "Saturday 10/08/2005 2:02:02 PM";
```

5- Observe the result:



So as you can see we cannot rely on the timestamp in “$standarn_Information” in Properties of any file. To check the real timing, we need to check Master File Table that will show the value of “$FileName” which reflects the real timestamp. You can check MFT from any Forensic tool such as Autopsy or FTKImager.

## Task 3: Prefetch Files:

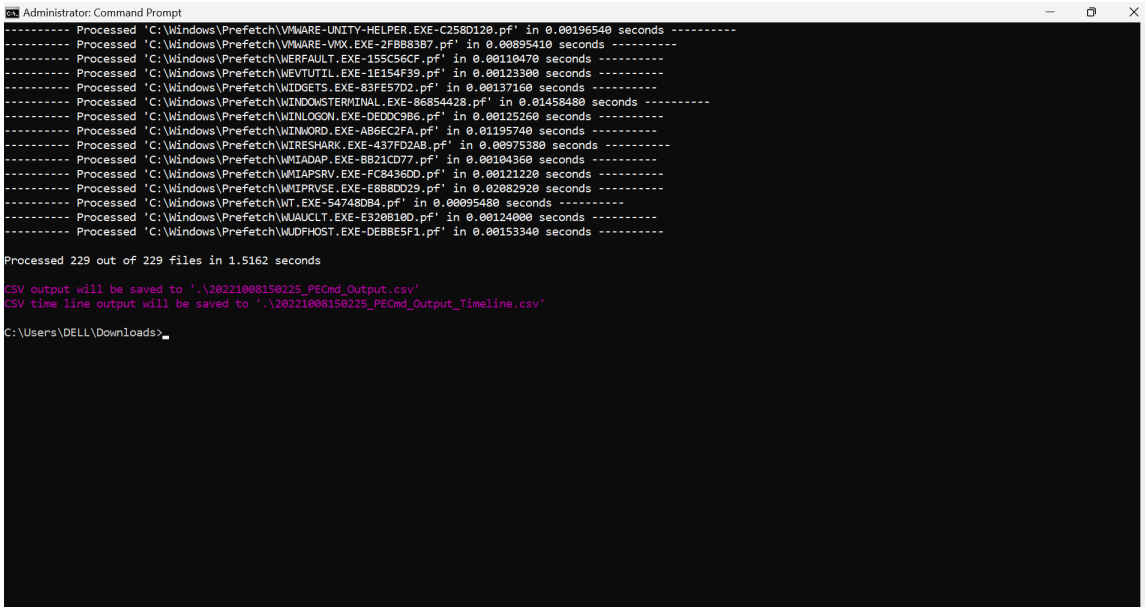
Prefetch files store information on applications that have been run on the system. Normally, it is in C:/Windows/Prefetch folder.

1- Download prefetch tool.
2- Open the terminal where Prefetch tool is installed. (Run Command Terminal AS ADMINISTRATOR)
3- Run the following command:

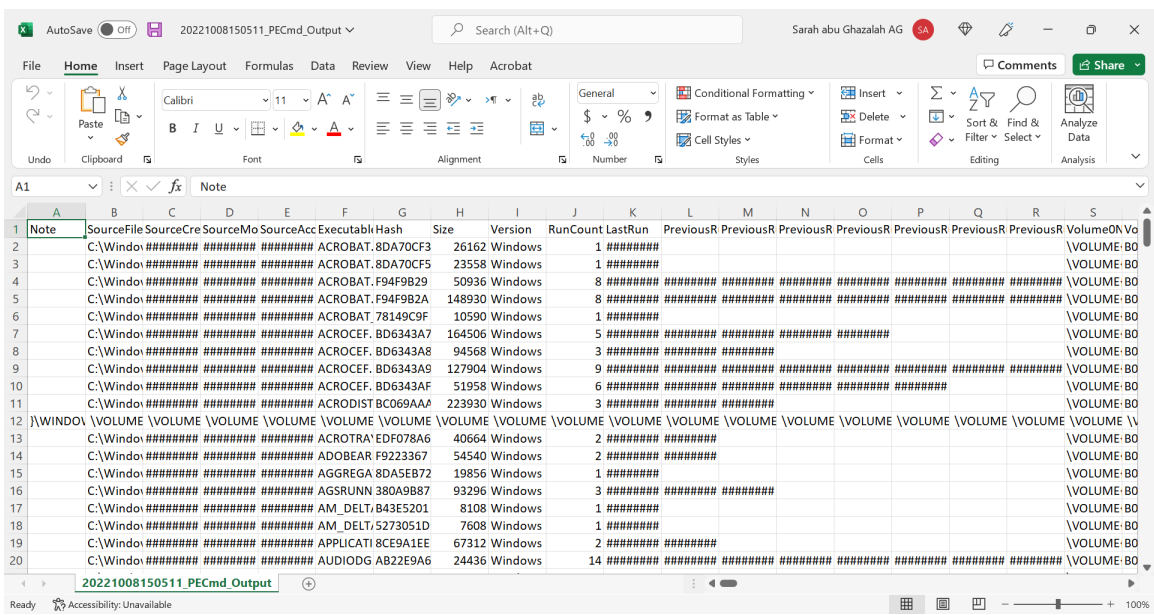
```
.\PECmd.exe -d C:\Windows\Prefetch\ --csv .\ -q
```

The command run Prefetch tool to output the contents of Prefetch to a csv file for analysis:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.521]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd /Users/DELL/Downloads

C:\Users\DELL\Downloads>.\PECmd.exe -d C:\Windows\Prefetch\ --csv .\ -q
PECmd version 1.4.0.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/PECmd

Command line: -d C:\Windows\Prefetch\ --csv .\ -q

Keywords: temp, tmp

Looking for prefetch files in 'C:\Windows\Prefetch\'


Found 229 Prefetch files

---------- Processed 'C:\Windows\Prefetch\ACROBAT.EXE-8DA70CF3.pf' in 0.02794590 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROBAT.EXE-8DA70CF5.pf' in 0.00070060 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROBAT.EXE-F94F9B29.pf' in 0.00123670 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROBAT.EXE-F94F9B2A.pf' in 0.00333050 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROBAT_SL.EXE-78149C9F.pf' in 0.00041040 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROCEF.EXE-BD6343A7.pf' in 0.00310800 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROCEF.EXE-BD6343A8.pf' in 0.00209680 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROCEF.EXE-BD6343A9.pf' in 0.00813880 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROCEF.EXE-BD6343AF.pf' in 0.00119600 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACRODIST.EXE-BC069AAA.pf' in 0.00372890 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ACROTRAY.EXE-EDF078A6.pf' in 0.00097410 seconds ----------
---------- Processed 'C:\Windows\Prefetch\ADOBEARM.EXE-F9223367.pf' in 0.00125810 seconds ----------
---------- Processed 'C:\Windows\Prefetch\AGGREGATORHOST.EXE-8DA5EB72.pf' in 0.00086030 seconds ----------
---------- Processed 'C:\Windows\Prefetch\AGSRUNNER.BIN-380A9B87.pf' in 0.00238110 seconds ----------
---------- Processed 'C:\Windows\Prefetch\AM_DELTA_PATCH_1.375.1668.0.E-B43E5201.pf' in 0.00040730 seconds ----------
---------- Processed 'C:\Windows\Prefetch\AM_DELTA_PATCH_1.375.1710.0.E-5273051D.pf' in 0.00036490 seconds ----------
---------- Processed 'C:\Windows\Prefetch\APPLICATIONFRAMEHOST.EXE-8CE9A1EE.pf' in 0.00752510 seconds ----------
---------- Processed 'C:\Windows\Prefetch\AUDIODG.EXE-AB22E9A6.pf' in 0.00074650 seconds ----------
---------- Processed 'C:\Windows\Prefetch\BACKGROUNDTASKHOST.EXE-6D58042C.pf' in 0.00113590 seconds ----------
---------- Processed 'C:\Windows\Prefetch\BACKGROUNDTRANSFERHOST.EXE-2A4FE575.pf' in 0.00144330 seconds ----------
```



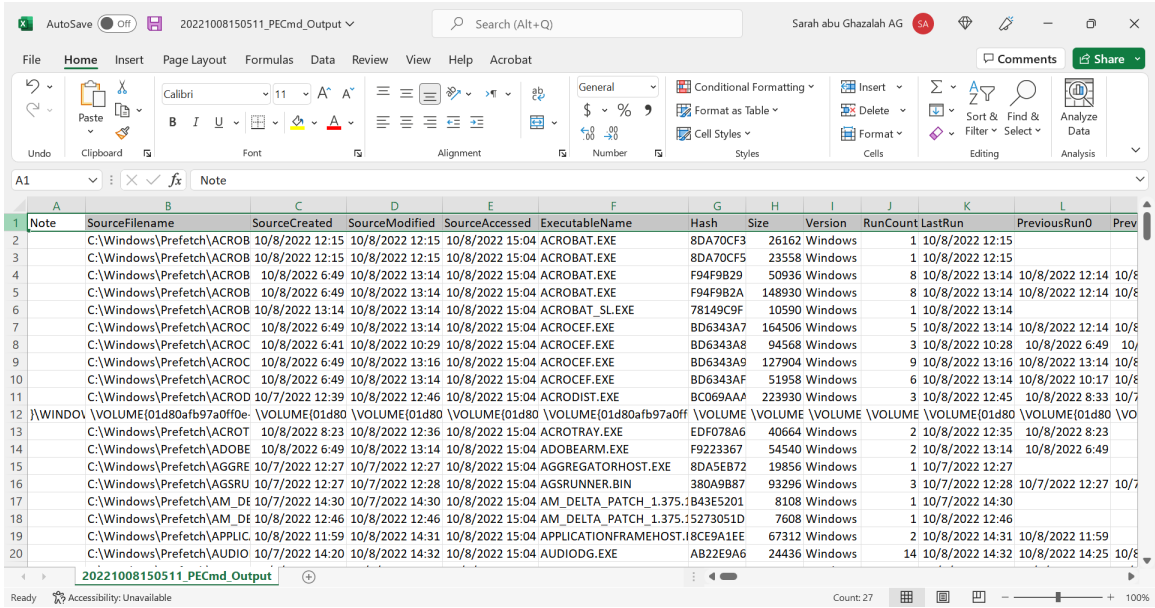

4- Open the csv file for the folder you are working from. (in my case Downloads).




Hint: Click on the hash values to display the digits correctly.

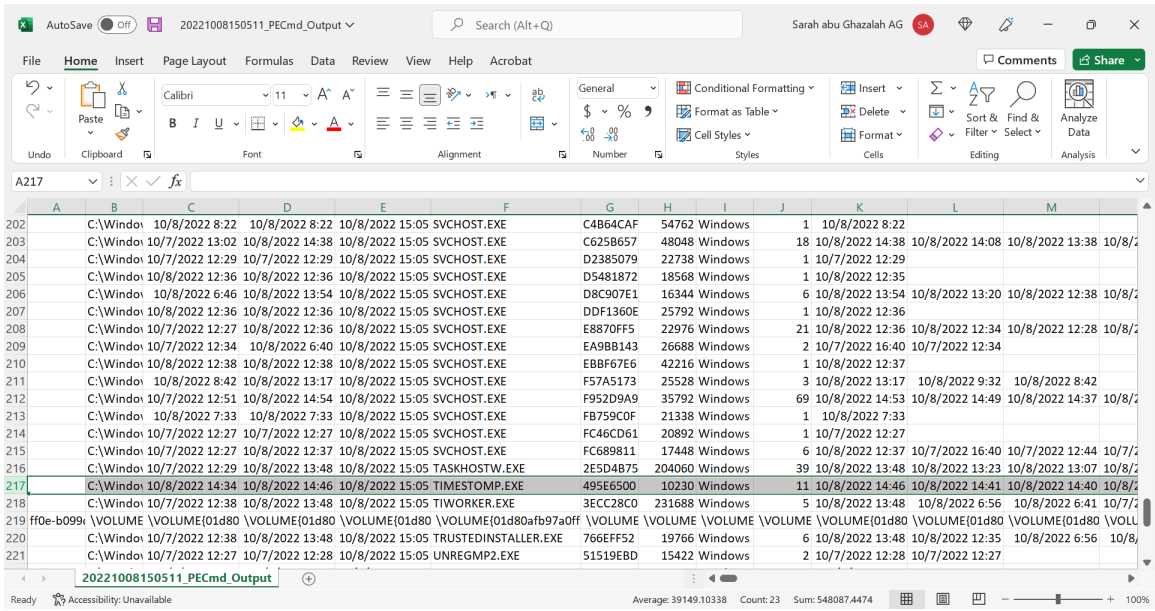
5- From the csv file you will find all the programs and processes run previously, even for programs you deleted before.

6- Check the headings in the file as shown in the image below, such as how many times the program was Run, dates and timing, Last Run, Previous Run (based on Run Count heading):



| | Note | SourceFilename | SourceCreated | SourceModified | SourceAccessed | ExecutableName | Hash | Size | Version | RunCount | LastRun | PreviousRun0 | Prev |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | | C:\Windows\Prefetch\ACROB | 10/8/2022 12:15 | 10/8/2022 12:15 | 10/8/2022 15:04 | ACROBAT.EXE | 8DA70CF3 | 26162 | Windows | 1 | 10/8/2022 12:15 | | |
| 3 | | C:\Windows\Prefetch\ACROB | 10/8/2022 12:15 | 10/8/2022 12:15 | 10/8/2022 15:04 | ACROBAT.EXE | 8DA70CF5 | 23558 | Windows | 1 | 10/8/2022 12:15 | | |
| 4 | | C:\Windows\Prefetch\ACROB | 10/8/2022 6:49 | 10/8/2022 13:14 | 10/8/2022 15:04 | ACROBAT.EXE | F94F9B29 | 50936 | Windows | 8 | 10/8/2022 13:14 | 10/8/2022 12:14 | 10/8 |
| 5 | | C:\Windows\Prefetch\ACROB | 10/8/2022 6:49 | 10/8/2022 13:14 | 10/8/2022 15:04 | ACROBAT.EXE | F94F9B2A | 148930 | Windows | 8 | 10/8/2022 13:14 | 10/8/2022 12:14 | 10/8 |
| 6 | | C:\Windows\Prefetch\ACROB | 10/8/2022 13:14 | 10/8/2022 13:14 | 10/8/2022 15:04 | ACROBAT_SL.EXE | 78149C9F | 10590 | Windows | 1 | 10/8/2022 13:14 | | |
| 7 | | C:\Windows\Prefetch\ACROC | 10/8/2022 6:49 | 10/8/2022 13:14 | 10/8/2022 15:04 | ACROCEF.EXE | BD6343A7 | 164506 | Windows | 5 | 10/8/2022 13:14 | 10/8/2022 12:14 | 10/8 |
| 8 | | C:\Windows\Prefetch\ACROC | 10/8/2022 6:41 | 10/8/2022 10:29 | 10/8/2022 15:04 | ACROCEF.EXE | BD6343A8 | 94568 | Windows | 3 | 10/8/2022 10:28 | 10/8/2022 6:49 | 10/ |
| 9 | | C:\Windows\Prefetch\ACROC | 10/8/2022 6:49 | 10/8/2022 13:16 | 10/8/2022 15:04 | ACROCEF.EXE | BD6343A9 | 127904 | Windows | 9 | 10/8/2022 13:16 | 10/8/2022 13:14 | 10/8 |
| 10 | | C:\Windows\Prefetch\ACROC | 10/8/2022 6:49 | 10/8/2022 13:14 | 10/8/2022 15:04 | ACROCEF.EXE | BD6343AF | 51958 | Windows | 6 | 10/8/2022 13:14 | 10/8/2022 10:17 | 10/8 |
| 11 | | C:\Windows\Prefetch\ACROD | 10/7/2022 12:39 | 10/8/2022 12:46 | 10/8/2022 15:04 | ACRODIST.EXE | BC069AAA | 223930 | Windows | 3 | 10/8/2022 12:45 | 10/8/2022 8:33 | 10/7 |
| 12 | }\WINDOV | \VOLUME{01d80afb97a0ff0e- | \VOLUME{01d80 | \VOLUME{01d80 | \VOLUME{01d80 | \VOLUME{01d80afb97a0ff | \VOLUME | \VOLUME | \VOLUME | \VOLUME | \VOLUME{01d80 | \VOLUME{01d80 | \VO |
| 13 | | C:\Windows\Prefetch\ACROT | 10/8/2022 8:23 | 10/8/2022 12:36 | 10/8/2022 15:04 | ACROTRAY.EXE | EDF078A6 | 40664 | Windows | 2 | 10/8/2022 12:35 | 10/8/2022 8:23 | |
| 14 | | C:\Windows\Prefetch\ADOBE | 10/8/2022 6:49 | 10/8/2022 13:14 | 10/8/2022 15:04 | ADOBEARM.EXE | F9223367 | 54540 | Windows | 2 | 10/8/2022 13:14 | 10/8/2022 6:49 | |
| 15 | | C:\Windows\Prefetch\AGGRE | 10/7/2022 12:27 | 10/7/2022 12:27 | 10/8/2022 15:04 | AGGREGATORHOST.EXE | 8DA5EB72 | 19856 | Windows | 1 | 10/7/2022 12:27 | | |
| 16 | | C:\Windows\Prefetch\AGSRU | 10/7/2022 12:27 | 10/7/2022 12:28 | 10/8/2022 15:04 | AGSRUNNER.BIN | 380A9B87 | 93296 | Windows | 3 | 10/7/2022 12:28 | 10/7/2022 12:27 | 10/7 |
| 17 | | C:\Windows\Prefetch\AM_DE | 10/7/2022 14:30 | 10/7/2022 14:30 | 10/8/2022 15:04 | AM_DELTA_PATCH_1.375.1 | B43E5201 | 8108 | Windows | 1 | 10/7/2022 14:30 | | |
| 18 | | C:\Windows\Prefetch\AM_DE | 10/8/2022 12:46 | 10/8/2022 12:46 | 10/8/2022 15:04 | AM_DELTA_PATCH_1.375.1 | 5273051D | 7608 | Windows | 1 | 10/8/2022 12:46 | | |
| 19 | | C:\Windows\Prefetch\APPLIC | 10/8/2022 11:59 | 10/8/2022 14:31 | 10/8/2022 15:04 | APPLICATIONFRAMEHOST.I | 8CE9A1EE | 67312 | Windows | 2 | 10/8/2022 14:31 | 10/8/2022 11:59 | |
| 20 | | C:\Windows\Prefetch\AUDIO | 10/7/2022 14:20 | 10/8/2022 14:32 | 10/8/2022 15:04 | AUDIODG.EXE | AB22E9A6 | 24436 | Windows | 14 | 10/8/2022 14:32 | 10/8/2022 14:25 | 10/8 |

7- For example, you will find timestomp tool in the csv file as shown below:



| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 202 | | C:\Windo | 10/8/2022 8:22 | 10/8/2022 8:22 | 10/8/2022 15:05 | SVCHOST.EXE | C4B64CAF | 54762 | Windows | 1 | 10/8/2022 8:22 | | |
| 203 | | C:\Windo | 10/7/2022 13:02 | 10/8/2022 14:38 | 10/8/2022 15:05 | SVCHOST.EXE | C625B657 | 48048 | Windows | 18 | 10/8/2022 14:38 | 10/8/2022 14:08 | 10/8/2022 13:38 | 10/8/2 |
| 204 | | C:\Windo | 10/7/2022 12:29 | 10/7/2022 12:29 | 10/8/2022 15:05 | SVCHOST.EXE | D2385079 | 22738 | Windows | 1 | 10/7/2022 12:29 | | |
| 205 | | C:\Windo | 10/8/2022 12:36 | 10/8/2022 12:36 | 10/8/2022 15:05 | SVCHOST.EXE | D5481872 | 18568 | Windows | 1 | 10/8/2022 12:35 | | |
| 206 | | C:\Windo | 10/8/2022 6:46 | 10/8/2022 13:54 | 10/8/2022 15:05 | SVCHOST.EXE | D8C907E1 | 16344 | Windows | 6 | 10/8/2022 13:54 | 10/8/2022 13:20 | 10/8/2022 12:38 | 10/8/2 |
| 207 | | C:\Windo | 10/8/2022 12:36 | 10/8/2022 12:36 | 10/8/2022 15:05 | SVCHOST.EXE | DDF1360E | 25792 | Windows | 1 | 10/8/2022 12:36 | | |
| 208 | | C:\Windo | 10/7/2022 12:27 | 10/8/2022 12:36 | 10/8/2022 15:05 | SVCHOST.EXE | E8870FF5 | 22976 | Windows | 21 | 10/8/2022 12:36 | 10/8/2022 12:34 | 10/8/2022 12:28 | 10/8/2 |
| 209 | | C:\Windo | 10/7/2022 12:34 | 10/8/2022 6:40 | 10/8/2022 15:05 | SVCHOST.EXE | EA9BB143 | 26688 | Windows | 2 | 10/7/2022 16:40 | 10/7/2022 12:34 | | |
| 210 | | C:\Windo | 10/8/2022 12:38 | 10/8/2022 12:38 | 10/8/2022 15:05 | SVCHOST.EXE | EBBF67E6 | 42216 | Windows | 1 | 10/8/2022 12:37 | | |
| 211 | | C:\Windo | 10/8/2022 8:42 | 10/8/2022 13:17 | 10/8/2022 15:05 | SVCHOST.EXE | F57A5173 | 25528 | Windows | 3 | 10/8/2022 13:17 | 10/8/2022 9:32 | 10/8/2022 8:42 | |
| 212 | | C:\Windo | 10/7/2022 12:51 | 10/8/2022 14:54 | 10/8/2022 15:05 | SVCHOST.EXE | F952D9A9 | 35792 | Windows | 69 | 10/8/2022 14:53 | 10/8/2022 14:49 | 10/8/2022 14:37 | 10/8/2 |
| 213 | | C:\Windo | 10/8/2022 7:33 | 10/8/2022 7:33 | 10/8/2022 15:05 | SVCHOST.EXE | FB759C0F | 21338 | Windows | 1 | 10/8/2022 7:33 | | |
| 214 | | C:\Windo | 10/7/2022 12:27 | 10/7/2022 12:27 | 10/8/2022 15:05 | SVCHOST.EXE | FC46CD61 | 20892 | Windows | 1 | 10/7/2022 12:27 | | |
| 215 | | C:\Windo | 10/7/2022 12:27 | 10/8/2022 12:37 | 10/8/2022 15:05 | SVCHOST.EXE | FC689811 | 17448 | Windows | 6 | 10/8/2022 12:37 | 10/7/2022 16:40 | 10/7/2022 12:44 | 10/7/2 |
| 216 | | C:\Windo | 10/7/2022 12:29 | 10/8/2022 13:48 | 10/8/2022 15:05 | TASKHOSTW.EXE | 2E5D4B75 | 204060 | Windows | 39 | 10/8/2022 13:48 | 10/8/2022 13:23 | 10/8/2022 13:07 | 10/8/2 |
| 217 | | C:\Windo | 10/8/2022 14:34 | 10/8/2022 14:46 | 10/8/2022 15:05 | TIMESTOMP.EXE | 495E6500 | 10230 | Windows | 11 | 10/8/2022 14:46 | 10/8/2022 14:41 | 10/8/2022 14:40 | 10/8/2 |
| 218 | | C:\Windo | 10/7/2022 12:38 | 10/8/2022 13:48 | 10/8/2022 15:05 | TIWORKER.EXE | 3ECC28C0 | 231688 | Windows | 5 | 10/8/2022 13:48 | 10/8/2022 6:56 | 10/8/2022 6:41 | 10/7/2 |
| 219 | ff0e-b099 | \VOLUME | \VOLUME{01d80 | \VOLUME{01d80 | \VOLUME{01d80 | \VOLUME{01d80afb97a0ff | \VOLUME | \VOLUME | \VOLUME | \VOLUME | \VOLUME{01d80 | \VOLUME{01d80 | \VOLUME{01d80 | \VOLU |
| 220 | | C:\Windo | 10/7/2022 12:38 | 10/8/2022 13:48 | 10/8/2022 15:05 | TRUSTEDINSTALLER.EXE | 766EFF52 | 19766 | Windows | 6 | 10/8/2022 13:48 | 10/8/2022 12:35 | 10/8/2022 6:56 | 10/8/ |
| 221 | | C:\Windo | 10/7/2022 12:27 | 10/7/2022 12:28 | 10/8/2022 15:05 | UNREGMP2.EXE | 51519EBD | 15422 | Windows | 2 | 10/7/2022 12:28 | 10/7/2022 12:27 | | |